

FY2022–FY2024 BIMO Form 483 Observation Taxonomy

Clinical Investigator Inspections — Seven Categories, Verified Citations, and Base-Rate Context

Purple Directive · Clinical Operations Practice

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FY2022–FY2024 BIMO Form 483 Observation Taxonomy

Clinical Investigator Inspections

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Methodology Note

This taxonomy was built from primary FDA sources, verified through direct access where the FDA's PDF servers allowed retrieval or through secondary analysis of those documents by legal and regulatory firms who confirmed the underlying data.

Primary sources consulted:

- FDA BIMO Inspection Metrics — FY2022 and FY2023. Official annual slide decks published by FDA. **FY2024 metrics have not been publicly released as of April 2026.** FDA typically publishes BIMO annual metrics 9–18 months after fiscal year close (FY ends September 30). FY2024 metrics should have been published in mid-to-late 2025, but as of this document's production date, no official FY2024 PDF is available.
- Goodwin Procter analysis of FDA BIMO violations (July 2023, July 2024), which directly cites the FDA metrics PDFs.
- FDA Warning Letters database — individual warning letters to clinical investigators.
- FDLI (Food and Drug Law Institute) compliance analysis (February 2025), which analyzed 42 BIMO-related warning letters from FY2019–FY2024.

- FDA Compliance Program Guidance Manual 7348.809 (Clinical Investigators).

Known limitations:

1. FDA does not publish percentage breakdowns of individual 483 observation categories at the 483 level. The metrics slides list observation categories but without item-level frequency percentages. Any percentage figures in this document are derived from warning letter analysis, not from the 483 database directly. These percentages reflect enforcement escalation, not raw 483 frequency.
 2. FY2024 inspection-level data (NAI/VAI/OAI counts, total inspections) is not yet published by FDA.
 3. The FDA's public Form 483 observation database (top-10 lists) covers drug manufacturers under GMP, not BIMO clinical investigator inspections. Those lists are a separate program. This is a common source of confusion.
 4. Verbatim warning letter text was retrieved through search engine indexing and secondary sources where direct FDA PDF access returned 404 errors. All verbatim quotes are confirmed authentic through multi-source verification.
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Part A: Top-Level Observation Categories

Category 1 — Protocol Non-Compliance / Failure to Follow the Investigational Plan

Primary 21 CFR citation: 21 CFR 312.60 — General Responsibilities of Investigators

“An investigator is responsible for ensuring that an investigation is conducted according to the signed investigator statement, the investigational plan, and applicable regulations.”

Additional citations: 21 CFR 312.53(c) (investigator qualifications); 21 CFR 812.100 (device IDE equivalent).

Frequency: This is the single most-cited basis for warning letter issuance to clinical investigators across FY2019–FY2024. Of 42 BIMO-related warning letters analyzed by FDLI over FY2019–FY2024, **25 (60%) cited 21 CFR 312.60** for failure to adhere to the investigational plan. It is also listed as the top observation category in both the FY2022 and FY2023 BIMO metrics. Goodwin Procter noted that in the first half of FY2024, four of six warning letters issued to clinical investigators involved this violation.

FDA does not publish an item-level percentage for Form 483 frequency; the 60% figure reflects warning letter escalation, not 483 frequency.

Top sub-patterns:

1a. Eligibility criteria violations. Enrolling subjects who do not meet inclusion criteria or who meet exclusion criteria, most commonly because screening assessments were not completed or documented before enrollment. This pattern appears in the Blanco (2023) and Goyal (2024) warning letters and is described as jeopardizing “*subject safety and welfare and rais[ing] concerns about the reliability of the data.*”

1b. Failure to perform protocol-required assessments or procedures. Omitting required safety imaging, labs, or clinical assessments; administering study treatment before required pre-dose safety checks; conducting assessments outside the required visit window without documentation. Han Phan (2024) received a warning letter because his “*failure to correctly administer and score the protocol-required assessments for the study raises significant concerns about the validity and integrity of the data collected at his site.*”

1c. Treatment administration errors linked to protocol. Dosing outside protocol parameters, administering investigational product before Quality Assurance release, or continuing treatment beyond the authorized period. The Angela Ritter (2024) warning letter cited administration of 50 doses beyond the authorized IND duration for individual patient expanded access under 21 CFR 312.305(c).

Warning letter example (verbatim-level):

- **WL:** 654873 — Robert J. Hayashi, M.D. / Washington University School of Medicine (March 14, 2023)
 - **Cited:** “*Failure to conduct the clinical study in accordance with the protocol resulted in the unnecessary administration of intrathecal methotrexate and the performance of invasive procedures, such as lumbar punctures and the sedation of a pediatric subject; and the initiation of study treatment cycles for a subject, despite that subject’s not meeting the treatment eligibility criterion.*”
 - 21 CFR 312.60
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Category 2 — Recordkeeping and Case History Inadequacies

Primary 21 CFR citations: 21 CFR 312.62(a) — Case histories (“*an investigator is required to prepare and maintain adequate and accurate case histories that record all observations and other data pertinent to the investigation on each individual administered the investigational drug*”); 21 CFR 312.62(b) — Record retention; 21 CFR 312.68 — Inspection of investigator’s records.

Frequency: Listed as the second most common observation category in both FY2022 and FY2023 BIMO metrics. The Goodwin analyses confirm this is a persistent top-two category. No item-level percentage is published by FDA for Form 483 frequency.

Top sub-patterns:

2a. Missing or incomplete source documents. Progress notes absent for study visits, clinical procedure records not in the subject file, no documentation of assessments that the protocol required, and missing records for adverse events that

were not captured in the CRF. The Clinical Pathways warning letter (2025, FY2024 inspection) cited: *“Multiple adverse events (AE) were not recorded”* and *“medical records not requested/reviewed for serious adverse events.”*

2b. Source data discrepancies and time-stamp inconsistencies. Conflicts between the sponsor’s CRF data, the site’s progress notes, and electronic system records for the same visit. The Clinical Pathways case documented *“time discrepancies between sponsor forms, progress notes, and electronic data.”* This pattern triggers data integrity concerns beyond the recordkeeping violation.

2c. Concomitant medication and prior history gaps. Concomitant medications not recorded per protocol timeline; prior medical history relevant to eligibility or safety not documented in the source record prior to enrollment. The Clinical Pathways case noted *“concomitant medication prior to transplant not recorded per protocol.”*

Warning letter example:

- **WL:** 668519 — Antonio E. Blanco, M.D. / Vista Health Research, LLC (September 26, 2023)
 - **Cited:** *“The enrollment of subjects who do not meet eligibility criteria, and the failure to perform protocol-required safety imaging procedures, jeopardized subject safety and welfare and raised concerns about the reliability of the data at his site.”*
 - 21 CFR 312.60 and 312.62
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Category 3 — Informed Consent Failures

Primary 21 CFR citations: 21 CFR 50.20 — General requirements for informed consent; 21 CFR 50.25(a) — Required elements of informed consent; 21 CFR 50.27 — Documentation of informed consent; 21 CFR 312.60 (general investigator responsibility encompasses consent process oversight).

Frequency: Listed as a top-five observation category in both FY2022 and FY2023 BIMO metrics. Goodwin’s 2024 analysis identifies it as a recurrent sub-pattern within the broader 312.60 violation cluster. It appears as a stand-alone citation (Part 50) and as a sub-element of 312.60 findings.

Top sub-patterns:

3a. Consent obtained by a non-delegated individual or by phone/remote method without documentation. The Goyal (2024) warning letter cited: *“progress notes indicated subjects were reconsented by phone, however no informed consent forms dated at the time of reconsent were located.”* The Clinical Pathways case cited *“third parties signed consent forms when participant signature required.”*

3b. Subjects enrolled or procedures performed before consent was obtained. This is the pattern with the most direct patient safety implications. Mark Dacey (2026, FY2025 inspection) received a warning letter for *“failure to obtain legally effective informed consent for a subject before collection of optional aqueous humor samples, despite the subject declining consent on multiple dates.”*

3c. Consent for protocol amendments not re-obtained. When a protocol amendment changes risk profile, a new consent is required. Sites commonly administer updated therapy or new procedures before obtaining updated consent documents, or use outdated consent forms after an approved amendment.

Warning letter example:

- **WL:** 660335 — Mobeen Mazhar, M.D. (May 31, 2023)
 - **Violation:** Failed to obtain informed consent in accordance with 21 CFR Part 50, cited under 21 CFR 312.60 and 21 CFR 50.20.
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Category 4 — Investigational Product Accountability Failures

Primary 21 CFR citations: 21 CFR 312.62(a) — Case histories (includes IP records); 21 CFR 312.57 (sponsor-level; applied at investigator sites when the investigator is also the sponsor); 21 CFR 312.60 (general responsibilities, encompasses IP control obligations under the signed Form FDA 1572).

Frequency: Listed as a top-five observation category in both FY2022 and FY2023 BIMO metrics. The FY2022 metrics and associated Goodwin analysis identify it as the fourth most commonly cited category.

Top sub-patterns:

4a. IP dispensed before Quality Assurance release. Documented in the Goyal (2024) warning letter: *“investigational product (IP) was administered before being released for infusion by the Quality Assurance Manager.”* This is a compound violation — protocol deviation plus IP control failure.

4b. Delegation failures for IP verification. IP accountability tasks performed by staff not listed on the delegation of authority log: *“IP verification was performed by a staff member who was not delegated the task on the site study log for investigator delegation.”*

4c. Dispensing records incomplete or inconsistent with drug accountability logs. Quantity dispensed, returned, and destroyed not reconciling with protocol supply tracking. This sub-pattern does not typically generate standalone warning letters but is a high-frequency 483 observation at the VAI level.

Category 5 — Adverse Event / Safety Reporting Failures

Primary 21 CFR citations: 21 CFR 312.64(b) — Investigator safety reports (*“an investigator shall promptly report to the sponsor any adverse effect that may reasonably be regarded as caused by, or probably caused by, the drug”*); 21 CFR 312.32 (IND safety reporting — sponsor-level but triggered by investigator’s failure to report to sponsor); 21 CFR 312.60 (general responsibilities).

Frequency: Listed as a top-five observation category in both FY2022 and FY2023 BIMO metrics. No item-level percentage is published. This category is more frequently cited on Form 483 than it generates standalone warning letters, because adverse event failures are often bundled with the broader 312.60 protocol non-compliance citation in warning letter issuance.

Top sub-patterns:

5a. AEs not recorded in source documents. Events documented verbally or in clinical notes but not captured in the CRF or on the AE log. Confirmed in the Clinical Pathways case: *“Multiple adverse events (AE) were not recorded.”*

5b. Failure to assess causality or relatedness. The investigator records an event but does not complete required causality assessment, leaving the sponsor unable to evaluate IND safety reporting obligations.

5c. SAEs not reported to the sponsor within the required timeframe. 21 CFR 312.64(b) requires prompt reporting. Delays — or sponsor notification only through routine CRF submission — appear across multiple inspections. The Clinical Pathways case documented that *“medical records not requested/reviewed for serious adverse events,”* meaning AEs were not even identified before inspection.

Category 6 — IRB Communication Failures

Primary 21 CFR citations: 21 CFR 312.66 — Assurance of IRB review (*“an investigator shall assure that an IRB that complies with the requirements set forth in part 56 will be responsible for the initial and continuing review and approval of the proposed clinical study. The investigator shall promptly report to the IRB all changes in research activity and all unanticipated problems involving risk to human subjects or others”*); 21 CFR 56.108 (IRB review of changes); 21 CFR 312.60 (general responsibilities).

Frequency: The FY2023 Goodwin analysis identified IRB-related violations as an increasing pattern in early FY2024, noting *“4 of 7 warning letters related to IRB activities”* in the FY2024 period observed through July 2024. This represents a notable shift from prior years. **Goodwin flags this as a trend to watch.**

Top sub-patterns:

6a. Protocol amendments or changes in research activity not reported to the IRB. The Ritter (2024) letter and Ralph DeFronzo (2025 letter, FY2025 inspection) both cite failure to promptly report changes in research activity to the IRB under 21 CFR 312.66. Extended drug administration beyond authorized duration — without amending the IND or notifying the IRB — is the most operationally common form of this violation.

6b. Research conducted under expired IRB approval. Lapsed continuing review, enrollment after approval expiration. Clinical operations teams commonly miss renewal deadlines because they rely on sponsor notification rather than maintaining their own IRB calendar.

6c. Unanticipated problems not reported to the IRB. Serious AEs or protocol deviations that should trigger IRB notification are not reported, either because the investigator did not recognize the reportability threshold or because communication broke down between the site coordinator and the PI.

Warning letter example:

- **WL:** 677324 — Namita A. Goyal, M.D. (October 10, 2024)
 - **Violation:** Failed to assure that an IRB complies with 21 CFR Part 56; failed to obtain IRB approval prior to changes in research activity, cited under 21 CFR 312.66. Compound violation spanning Categories 1, 4, and 6.
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Category 7 — Failure to Submit IND Application (Sponsor-Investigator Violations)

Primary 21 CFR citation: 21 CFR 312.20(a) — Requirement for IND (“a sponsor shall submit an IND to FDA if the sponsor intends to conduct a clinical investigation”); 21 CFR 312.2(a) (IND exemption criteria, which investigators frequently misapply).

Frequency: Of 42 BIMO-related warning letters analyzed from FY2019–FY2024, **13 (31%) cited failure to submit an IND**, making this the second most common basis for warning letter issuance. 7 of 13 involved sponsor-investigators who were conducting investigations with unapproved drugs and either believed an exemption applied or were unaware of the requirement.

Top sub-patterns:

7a. Sponsor-investigator misapplication of IND exemptions. Investigators using marketed drugs for off-label purposes in research settings, or combining approved drugs in novel combinations, without recognizing that an IND is required when the research is intended to support a label change or when subjects will be exposed to unreasonable risk.

7b. Lack of awareness that investigator-initiated trials require sponsor-investigator IND. Particularly common in academic settings and community oncology practices. The investigator treats the protocol as a practice-improvement initiative rather than a clinical investigation under 21 CFR 312.

7c. Amended protocols not reflected in IND updates. Study conducted under conditions (dose, duration, indication) that deviate materially from the submitted IND without filing a protocol amendment. The Ritter (2024) warning letter exemplifies this: 50 additional doses administered beyond authorized IND parameters with no protocol amendment filed.

Part B: Base-Rate Context

Inspection Volume: FY2022 and FY2023

Metric	FY2022	FY2023	FY2024
Total BIMO inspections (all categories)	766	1,073	<i>Not published</i>
Clinical investigator inspections (total)	504	681	<i>Not published</i>
CI inspections — drug/bio/device Centers	468	666	<i>Not published</i>
NAI (No Action Indicated)	408	536	<i>Not published</i>
VAI (Voluntary Action Indicated)	87	136	<i>Not published</i>
OAI (Official Action Indicated)	9	9	<i>Not published</i>
Warning letters following BIMO	~6	~11	~12

Notes on this table:

- The FY2022 and FY2023 warning letter counts are reported as a combined category by Goodwin; not all are issued to the clinical investigator — some are issued to the sponsoring entity when the inspection was of a sponsor/CRO. The FY2024 count of 12 is from the FDLI analysis of FY2024 warning letters.
- OAI count of 9 is identical in FY2022 and FY2023. This is not a typographic error — it is the number reported in both official BIMO metrics publications.
- FY2024 inspection-level data has not been published. As of April 2026, FDA has released FY2023 as the most recent complete BIMO metrics report.

Most-Cited 21 CFR Sections — Warning Letter Frequency (FY2019–FY2024)

Rank	Section	Subject	Warning Letters (of 42)
1	21 CFR 312.60	General investigator responsibilities / protocol compliance	25 (60%)
2	21 CFR 312.20	Failure to submit IND	13 (31%)
3	21 CFR 50.20 / Part 50	Informed consent	Frequent; co-cited with 312.60
4	21 CFR 312.66	Assurance of IRB review	Increasing; 4 of 7 in early FY2024
5	21 CFR 312.62	Investigator recordkeeping	High 483 frequency; less common as standalone WL

Source: FDLI analysis of 42 BIMO warning letters FY2019–FY2024 (February 2025).

21 CFR 312.62 generates a high volume of Form 483 observations at the site level but less frequently rises to standalone warning letter issuance. It typically co-appears with 312.60 violations.

Trend Notes Across FY2022–FY2024

- Inspection volume increased sharply from FY2022 to FY2023.** Total BIMO inspections rose from 766 to 1,073 (+40%), and clinical investigator inspections rose from 504 to 681 (+35%). This reflects post-pandemic recovery of the inspection program. OAI rates remained low (9 per year both FY2022 and FY2023), indicating FDA is not seeing a systemic quality collapse at sites — most findings are VAI-level.
- Warning letter volume is increasing.** Approximately 6 in FY2022, 11 in FY2023, 12 in FY2024. The upward trajectory tracks with increased inspection activity.
- IRB-related violations are a growing enforcement focus.** Goodwin’s mid-FY2024 analysis specifically flagged this trend. 4 of 7 BIMO warning letters in the early FY2024 period involved IRB activities.
- IND submission failures are the second most common warning letter basis** and disproportionately affect sponsor-investigators (physician-initiated research). This reflects FDA’s continued enforcement attention on academic and community site investigators who self-sponsor protocols without adequate regulatory infrastructure.

5. **Data integrity findings are escalating in severity.** The Applied Therapeutics case (2024 warning letter) resulted in a concurrent Complete Response Letter (CRL) to the drug application and a shareholder lawsuit following FDA’s BIMO finding that electronic records were deleted two days after pre-announced inspection.

6. **FY2024 full metrics are not yet public.** Treat FY2024 as directional data only.

Part C: Regulatory Reference Map

Key regulatory framework citations for clinical investigator obligations. These are the sections that drive the observation categories above.

Section	Title	Operational Requirement
21 CFR 312.60	General responsibilities of investigators	Ensure study conducted per investigational plan; protect rights and welfare of subjects; control investigational drug
21 CFR 312.62(a)	Investigator recordkeeping — case histories	Prepare and maintain adequate and accurate case histories for each subject
21 CFR 312.62(b)	Investigator recordkeeping — retention	Retain records for 2 years after NDA approval or study discontinuation
21 CFR 312.64(b)	Investigator safety reports	Promptly report adverse effects to sponsor
21 CFR 312.66	Assurance of IRB review	Ensure IRB-compliant review; promptly report changes and unanticipated problems
21 CFR 312.68	Inspection of investigator’s records	Permit FDA access to and copying of all records maintained under 312.62
21 CFR 312.20(a)	IND requirement	Sponsor must submit IND before commencing clinical investigation
21 CFR 50.20	General informed consent requirement	Legally effective informed consent required from each subject
21 CFR 50.25(a)	Elements of informed consent	Eight required disclosure elements
21 CFR 50.27		

Section	Title	Operational Requirement
	Documentation of informed consent	Written consent required; signed and dated by subject

All sections above are accessible at [ecfr.gov/current/title-21](https://www.ecfr.gov/current/title-21).

What This Document Does Not Claim

This taxonomy does not assign percentage frequencies to individual Form 483 observation categories at the 483 level (e.g., “protocol non-compliance represents 35% of all clinical investigator 483s”). FDA does not publish that breakdown for the BIMO program. The frequencies cited in this document reflect warning letter issuance — a post-483, post-EIR enforcement action — not raw 483 frequency. Warning letters represent a small subset of OAI-classified inspections and an even smaller subset of all inspections.

The pattern in VAI inspections (which constitute the majority of findings — 87 in FY2022, 136 in FY2023) is not publicly documented at the category level. The categories identified in the BIMO metrics slides represent qualitative groupings confirmed by FDA; their relative frequency at the VAI level remains unpublished.

Source Citations

1. Food and Drug Law Institute (FDLI). “*Compliance Challenges for Clinical Research Sites*.” February 2025. Analysis of 42 BIMO-related warning letters from FY2019–FY2024. <https://www.fdpi.org/2025/02/compliance-challenges-for-clinical-research-sites/>
2. Goodwin Procter LLP. “*Common FDA Bioresearch Monitoring (BIMO) Violations: Updates from FY 2023 to Now*.” July 2024. <https://www.goodwinlaw.com/en/insights/publications/2024/07/alerts-lifesciences-common-fda-bioresearch-monitoring-violations>
3. FDA BIMO Inspection Metrics — FY2022 (<https://www.fda.gov/media/165853/download>) and FY2023 (<https://www.fda.gov/media/178661/download>). Index page: <https://www.fda.gov/science-research/clinical-trials-and-human-subject-protection/bimo-inspection-metrics>
4. FDA Warning Letter 654873 — Robert J. Hayashi, M.D. (March 14, 2023). <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/robert-j-hayashi-mdwashington-university-school-medicine-department-pediatrics-654873-03142023>

5. FDA Warning Letter 668519 — Antonio E. Blanco, M.D. (September 26, 2023). <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/warning-letters/antonio-e-blanco-mdvista-health-research-llc-668519-09262023>
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10. Cooley LLP. “Beware of BIMO: Complete Response Letter, Warning Letter and Shareholder Lawsuit Follow FDA Data Integrity Findings.” January 14, 2025. <https://www.cooley.com/news/insight/2025/2025-01-14-beware-of-bimo-complete-response-letter-warning-letter-and-shareholder-lawsuit-follow-fda-data-integrity-findings>
11. Clinical Pathways Research. “FDA BIMO Inspection Results in Warning Letter.” February 19, 2025. <https://www.clinicalpathwaysresearch.com/blog/2025/2/19/fda-bimo-inspection-results-in-warning-letter>

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